

1 • Shared evidence: change detectives

Reversible change detectives

Name _____ Date _____

Records C and D are constructed examples. All work can be completed without a practical.

A • Ice record

Solid ice changes to liquid water when energy is transferred in. Later, enough cooling turns the liquid back into ice.

C • Mass record

Constructed data: sealed container and contents are 120 g before dissolving and 120 g after. Nothing is added or removed while dissolving.

E • Distinction

Melting is a solid-to-liquid state change. Dissolving mixes a substance through a solvent; a clear solution can still contain the dissolved substance.

B • Sugar record

Sugar crystals are stirred into room-temperature water. The crystals are no longer visible; the liquid stays clear. No tasting is used.

D • Recovery record

Constructed observation over several days: water leaves an open sugar solution by evaporation; dry sugar remains. No class trial or heating took place.

F • Boundary

The diagrams model the stated physical examples. “Easy to reverse” alone is not a complete test for whether any unknown change is physical or chemical.

W1. Name the processes in A and B.

W2. What do the mass and recovery records show about sugar?

W3. Repair “Dissolving makes sugar into water”.

2 • Supported route: choose the process

Use page 1. Word bank: melting / freezing / dissolving / solution.

S1. Ice becomes liquid water. This is ...

- A. Melting
- B. Dissolving

S2. Liquid water becomes ice. This is ...

- A. Freezing
- B. Evaporation

S3. Sugar mixes through water. This is ...

- A. Dissolving
- B. Melting

S4. Explain where the sugar is after crystals are no longer visible.

S5. Choose how to recover sugar: remove water by evaporation / pretend it vanished. Explain.

S6. Does open evaporation collect the water as well?

3 • Standard route: process report

Track the starting substance and the result.

Process	Starting situation	Result
Melting		
Freezing		
Dissolving sugar		

N1. Complete the table with the lesson examples.

N2. Explain the difference between melting and dissolving.

N3. Use both C and D to challenge “the sugar no longer exists”.

N4. Describe how to recover sugar and state which component open evaporation does not collect.

4 • Stretch route: evidence and recovery

Explain the limits as well as the useful result.

T1. Compare melting ice and dissolving sugar using substances, states and particle arrangements.

T2. Does a constant mass of 120 g alone prove exactly what substances are present? Explain why recovery evidence adds information.

T3. Evaluate “Cooling any mixture always returns all starting materials separately”. Use the two cases.

T4. Explain why recovering sugar by open evaporation is different from recovering both sugar and water.

5 • Exit and reflection

Use accurate process words.

E1. How is melting different from dissolving?

E2. Where is the sugar in a clear sugar solution?

E3. How can sugar be recovered in this example?

E4. What process reverses melting of water?

Reflection. Which change would need a different recovery method?
